- ***This rubric explains how the quality of a student's Assignment 2 submission is evaluated.***
- ***If required artefacts (tables, plots, or rule matrix) specified in each task are missing or not presented in the specified structure, the mark awarded for the corresponding criterion will be reduced accordingly.***
- ***For each criterion, markers select one level that best reflects the overall quality of the submission and award a mark within the indicated range.***

SIT215 Assignment 2: Intelligent System Design with Fuzzy Logic – Rubric

Criteria	Developing	Nailing it	Solid work	Exceptional quality
Problem Definition and Fuzzy Variables (4)	Scenario unclear or variables poorly chosen. Mandatory variable missing or linguistic levels fewer than three. Numeric ranges are inappropriate or unjustified. (0-1)	Scenario defined, but variable relationships are weak or partially inconsistent with the decision problem. Linguistic labels present but limited justification. (2)	Scenario appropriate with logically related variables. At least three linguistic levels per variable and reasonable numeric ranges. Contextual justification is mostly clear. (3)	Scenario is clearly defined and strongly relevant to the delivery context. Variables are well chosen, ranges are justified, and linguistic labels are meaningful. Deep contextual justification for all ranges is demonstrated. (4)
Membership Function Design (6)	Membership functions are poorly designed or inconsistent. Overlap is missing or not clearly annotated. (0-1)	Membership functions present with correct plots and annotations. Membership functions are correct, but the overlap explanation is limited, or the graphs lack professional clarity. (2-3)	Membership functions are presented, with correct plots and clear annotations. Membership functions are correct, with a reasonable explanation of overlap. Minor annotation gaps are observed. (4-5)	Membership functions are clearly designed for all variables. Plots show professional precision with clear, annotated overlap. Design supports fuzzy reasoning with clear justification for smooth transition and boundary logic. (6)
Fuzzy Rule Base Design (6)	The rule matrix is unclear, or the rules are weakly connected to the variables. (0-1)	The rule matrix is present with connected variables, but the rules are simplistic or partially justified. (2-3)	The rule matrix is present with connected variables. The rule base demonstrates coherent and logical rule relationships. (4-5)	A logically robust rule matrix is constructed. A full set of meaningful, aligned fuzzy rules is present. Clear trade-offs between input variables demonstrate strong fuzzy reasoning. (6)
Reasoning Demonstration and Explanation (4)	The worked example is missing or poorly structured. Input values are unclear or the reasoning stages of the fuzzy system are not fully demonstrated. (0-1)	An example of a meaningful, ambiguous case is present, but some key stages of fuzzy reasoning (such as fuzzification, rule activation, inference, or defuzzification) are weakly explained or partially missing. (2)	The worked example demonstrates most reasoning stages, including fuzzification, rule activation, and output reasoning. The explanation and evaluation conclusion are mostly clear, though some reasoning steps or justification may lack depth. (3)	A well-chosen ambiguous example clearly demonstrates the full reasoning process. Fuzzification, rule activation, inference, and final output, as well as the conclusive justification, are logically explained, and the resulting decision is consistent with the system design. (4)